

Evaluation Design Determines Reported Performance in Actigraphy-Based Machine
Learning for Mental Disorders:
A PROBAST+AI/TRIPOD+AI Appraisal of Published Models and a Controlled
Re-evaluation of Subject-wise Validation,
Cross-cohort Transfer and Calibration on DEPRESJON, PSYKOSE and HYPERAKTIV

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Code, run manifests and generated tables: <https://github.com/rogerpanel/cv/tree/main/mlmh-appraisal>

Abstract

Background and objective. The open actigraphy cohorts DEPRESJON (depression), PSYKOSE (schizophrenia) and HYPERAKTIV (ADHD), harmonised in 2025 as OBF-Psychiatric, are the most used public benchmarks for machine-learning (ML) detection of mental disorders from wearables. Reported accuracies have risen from the dataset authors' leave-one-patient-out 0.7 to above 0.9. We asked whether that reflects better models or weaker evaluation, and measured what the prevalent evaluation practices cost. **Methods.** Part I appraises every published supervised model on the four cohorts with PROBAST+AI and TRIPOD+AI, extracting the unit of train/test splitting, external validation, calibration reporting, events per predictor, code availability, and whether the control group shared by DEPRESJON and PSYKOSE was double counted. Part II re-evaluates the cohorts with models fixed a priori (logistic regression, random forest, XGBoost, 1D-CNN), ten seeds and subject-level bias-corrected bootstrap intervals, in six experiments: E1 record-wise versus subject-wise cross-validation on identical data; E2 frozen cross-cohort transfer with shared controls assigned to one cohort; E3 calibration (slope, intercept, Brier, ECE, reliability curves, recalibration); E4 sex strata; E5 transdiagnostic and five-class tasks on OBF-Psychiatric; E6 feature-group, resampling and cross-validation-scheme ablation. **Results.** Of 28 published models identified before the formal search, 7 split by participant, 9 by day or window and 12 did not state the unit; none reported calibration or validated on a second cohort; 4 released code. Record-wise splitting inflated window-level AUROC by 0.065 to 0.114 on DEPRESJON, 0.020 to 0.044 on PSYKOSE and 0.130 to 0.169 on HYPERAKTIV, where the honest estimate was at chance (0.44 to 0.45) and the leaky one reached 0.60 to 0.65 at participant level. Models trained on PSYKOSE lost 0.13 to 0.21 AUROC on DEPRESJON with calibration slopes of 0.27 to 0.63; honest slopes were 0.53 to 0.95 and ECE 0.06 to 0.22; recalibration from the training cohort did not repair the transfer. Participant-level events per predictor were 0.6 to 0.8. Honest performance was stable across feature subsets and cross-validation schemes (E6). **Conclusions.** Reported progress on these benchmarks is largely an artefact of evaluation design. We provide subject-wise, calibrated, cross-cohort baselines with public code, manifests and a TRIPOD+AI self-audit against which future claims on these cohorts can be checked.

Keywords: actigraphy; depression; schizophrenia; ADHD; data leakage; external validation; calibration; PROBAST+AI; TRIPOD+AI; reproducibility.

1 Introduction

1.1 Background

Wearable actigraphy records motor activity continuously and unobtrusively, and disturbed activity and circadian rhythm are among the best-replicated objective correlates of depression, psychosis and ADHD. Three open datasets from Haukeland University Hospital and collaborating Norwegian sites made this signal available to the ML community: DEPRESJON (23 depressed in- and outpatients, 32 healthy controls) [1], PSYKOSE (22 inpatients with schizophrenia and the same 32 controls) [2] and HYPERAKTIV (adults with ADHD and clinical controls) [3], all recorded with one device family at one-minute resolution and re-released in 2025 as the harmonised OBF-Psychiatric collection of 162 participants [4]. They are now the default benchmarks for detecting mental disorders from wearables, and the reported numbers have climbed: the dataset authors’ leave-one-patient-out F1 of 0.73 for depression [1] has been followed by accuracies of 0.85 to 0.96 [5, 6, 7] and, for schizophrenia, of 0.92 to 0.94 [8, 9].

1.2 The evaluation problem

Each participant contributes many days of data. If days are shuffled into cross-validation folds regardless of participant (*record-wise* splitting), a flexible model can recognise the person rather than the disorder, and performance is inflated; this is documented in wearable, EEG and neuroimaging research [10, 11] but has not been quantified on these cohorts. Two further practices compound it. Models are almost never applied to a second cohort, so the loss of discrimination and calibration under population shift is unknown. And calibration, the agreement between predicted probability and observed frequency, is essentially never reported, although a screening tool is used through its probabilities and not its rank order [12]. The PROBAST+AI risk-of-bias tool [13] and the TRIPOD+AI reporting statement [14] now make these explicit requirements for ML prediction models; a field-wide appraisal of psychiatric prediction models found calibration reported in 22% and adequate internal validation in 11% [15]. A dataset-specific hazard adds to the general ones: the PSYKOSE control group is the DEPRESJON control group, so any study that combines the two cohorts, or trains on one and tests on the other, without recognising the overlap has scored part of its test set on training participants.

1.3 Gaps and contributions

No study has (i) appraised the published models on these cohorts systematically, (ii) measured the paired difference between record-wise and subject-wise evaluation on the same data, models and seeds, (iii) reported cross-cohort transfer with the shared controls handled, or (iv) reported calibration for any model on these cohorts. This paper does all four, in one design (Figure 1), and contributes:

1. a closed-scope systematic appraisal (Part I) with PROBAST+AI and TRIPOD+AI, whose outcomes are chosen so that each can be priced on the data (Section 2);
2. a controlled re-evaluation (Part II) that quantifies leakage inflation (E1), cross-cohort loss with calibration collapse (E2), the calibration blind spot with a recalibration test (E3), sex-stratified performance (E4), transdiagnostic and five-class tasks on the de-duplicated 162-participant release (E5), and an ablation over feature groups, resampling and cross-validation scheme (E6), with subject-level bootstrap uncertainty throughout;
3. a comparison of every published claim against the measured honest and leaky bands (Table 10, Figure 2), a calibration comparison (Figure 5) and a multi-metric profile of the models (Figure 6);
4. a public, tested codebase in which subject-wise splitting is enforced by a build-failing test, cross-cohort overlap is detected from the data, every run writes a manifest, and every table

in this paper is generated from those manifests, together with a TRIPOD+AI self-audit.

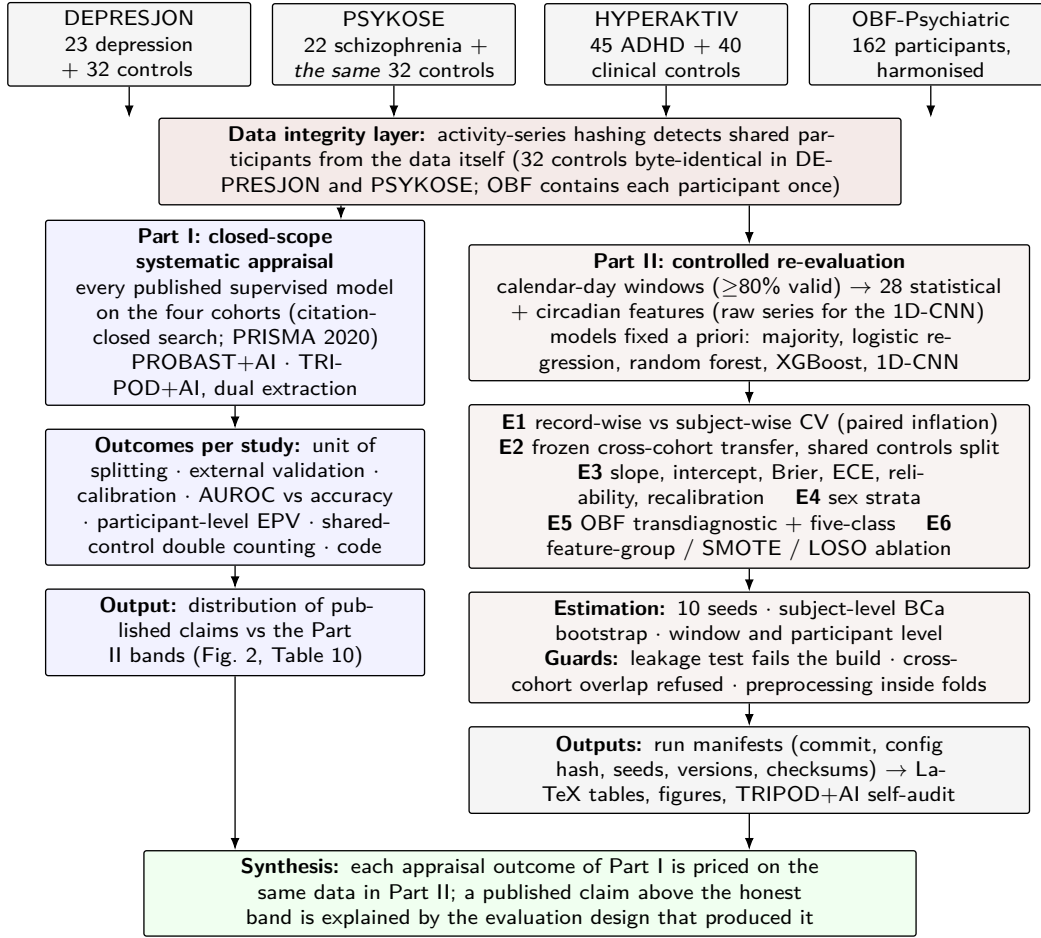


Figure 1: Architecture of the study. Four open actigraphy cohorts pass through a data-level integrity check (participant hashing) into a closed-scope systematic appraisal of the published models (Part I) and a controlled re-evaluation under the designs the appraisal scores (Part II); the synthesis maps every appraisal outcome to a measured cost.

2 Part I: systematic appraisal of the published models

2.1 Design

Part I is a closed-scope systematic review registered [PROSPERO/OSF ID] and reported per PRISMA 2020. *Eligibility*: peer-reviewed articles, indexed conference papers and preprints (2018 to [search date], English) that train or evaluate a supervised model to predict group membership, symptom score or state from DEPRESJON, PSYKOSE, HYPERAKTIV or OBF-Psychiatric activity data and report at least one performance metric; excluded are reviews, dataset descriptors without a model, and unsupervised or purely descriptive analyses. *Search*: forward citation chasing of the four dataset papers in Scopus, Web of Science and Google Scholar, plus the dataset names as search terms in PubMed, Scopus, Web of Science, IEEE Xplore and the ACM Digital Library (Supplementary Table S3 records counts and dates; [to be run]). *Screening and extraction*: two reviewers independently (Rayyan; Cohen’s κ), disagreements by discussion. *Appraisal*: PROBAST+AI (four domains) and the 27-item TRIPOD+AI checklist in duplicate. *Pre-specified outcomes per study*: unit of splitting (participant, day/window, unstated); any external validation; calibration reported; AUROC reported or accuracy only; participant-level events per candidate predictor; use of both DEPRESJON and PSYKOSE and, if so, recognition of the shared

control group; code availability; and the best reported performance, so that claims can be plotted against the Part II bands.

2.2 Seed extraction and provisional results

Before the formal search, 28 studies were identified from the dataset papers’ citation trails and topic searches (Table 1; sources in Supplementary Table S4). This seed set is not the review’s final sample, and the cells extracted from abstracts alone are marked in red for verification against full texts during dual extraction; the pattern is nonetheless already clear. Of the 28, 7 split by participant (the four dataset papers and three follow-ups), 9 split by day or window, and 12 did not state the unit in the abstract. None reported any calibration measure. None validated a model on a second cohort; one preprint claims an independent set under a “modified leave-one-out” scheme. Four released code, all of them dataset papers. Every study reporting accuracy above 0.85 on DEPRESJON or above 0.90 on PSYKOSE used, or did not state, a day-level unit. Four studies combined DEPRESJON and PSYKOSE; [how many recognised the shared controls: to be confirmed at full-text stage; none states it in the abstract]. Participant-level events per predictor could be computed for none from the abstract and will be extracted from full texts; for the cohorts themselves the value is bounded above by 23 cases divided by the number of candidate predictors, i.e. below 1 for any feature set larger than 23.

Table 1: Part I seed extraction: published supervised models on the four cohorts identified before the formal search (Supplementary Table S4 lists sources). Red cells are extractions from abstracts that must be verified against the full text during dual extraction; the formal search will add studies to this list.

ID	First author	Year	Cohorts	Best model	Split unit	Best reported	Calib.	Ext. val.	Code
S01	Garcia-Ceja	2018	DEPRESJON	Random forest / DNN	participant (leave-one-patient-out)	F1 0.73	no	no	yes
S02	Zanella-Calzada	2019	DEPRESJON	Random forest	day/window	accuracy 0.89	no	no	no
S03	Frogner	2019	DEPRESJON	1D-CNN	participant (leave-one-participant-out)	F1 verify	no	no	no
S04	Jakobsen	2020	DEPRESJON	Random forest / DNN / CNN	participant (leave-one-out)	accuracy verify	no	no	no
S05	Jakobsen	2020	PSYKOSE	Classical ML	participant (leave-one-patient-out)	F1 verify	no	no	yes
S06	Hicks	2021	HYPERAKTIV	Classical ML	untested (10-fold on features)	accuracy verify	no	no	yes
S07	Rodriguez-Ruiz	2022	DEPRESJON	2D-CNN	day/window	accuracy 0.767	no	no	no
S08	Rodriguez-Ruiz	2022	DEPRESJON+PSYKOSE	CNN	day/window	accuracy verify	no	no	no
S09	Kumar	2022	DEPRESJON	ML on autoregressive features	untested	accuracy 0.78	no	no	no
S10	Ahmed (arXiv)	2023	DEPRESJON	Transfer learning ensemble	modified leave-one-out (verify)	accuracy 0.96	no	claims independent set	no
S11	Hybrid RF-ANN (arXiv)	2023	DEPRESJON	RF-ANN ensemble	untested	accuracy verify	no	no	no
S12	UMAP study	2023	DEPRESJON	ML + UMAP	untested	accuracy verify	no	no	no
S13	Genetic-algorithm feature selection	2023	DEPRESJON	GA feature selection + classifier	day/window (intervals)	accuracy verify	no	no	no
S14	Attention CNN (IoMT)	2023	PSYKOSE	Deep CNN + attention	day/window	accuracy verify	no	no	no
S15	Multi-branch DL	2024	PSYKOSE	Multi-branch deep network	day/window	accuracy 0.92	no	no	no
S16	DL + XAI	2024	PSYKOSE (+DEPRESJON)	Deep network + XAI	day/window (24-h segments)	accuracy 0.94	no	no	no
S17	Hopping-mean augmentation	2024	DEPRESJON	Augmented ML	day/window	accuracy verify	no	no	no
S18	IoMT dimensionality reduction	2024	HYPERAKTIV	10 classifiers on reduced features	untested	accuracy verify	no	no	no
S19	Explainable XGBoost	2025	DEPRESJON	XGBoost + ADASYN	untested in abstract (verify)	accuracy 0.849	no	no	no
S20	Improved CNN framework	2025	DEPRESJON+PSYKOSE	Improved CNN	day/window (verify)	accuracy verify	no	no	no
S21	Enhanced ADHD ML	2025	HYPERAKTIV/OBF (45 participants)	SVM	untested (verify)	F1 0.779	no	no	no
S22	HYPERAKTIV ML study	2025	HYPERAKTIV	RF / LightGBM / SVM	untested	accuracy verify	no	no	no
S23	Multimodal digital biomarkers	2025	HYPERAKTIV	Bernoulli NB (actigraphy+HRV)	untested (verify)	F1 0.79	no	no	no
S24	OBF-Psychiatric descriptor	2025	OBF-Psychiatric	Classical ML baselines	participant (verify)	accuracy verify	no	no	yes
S25	Data-uncertainty case study	2025	OBF-Psychiatric (ADHD)	QC + ML	participant (verify)	nan no difference after QC	no	no	no
S26	VIT / CoAtNet on images	2025	DEPRESJON+PSYKOSE	CoAtNet-Tiny	participant (three-fold subject-wise)	accuracy verify	no	no	no
S27	Time-segmentation evaluation	2026	DEPRESJON	ML on day segments	verify	accuracy verify	no	no	no
S28	Accurate ADHD identification	2022	HYPERAKTIV	ML on real-time activity	untested (verify)	accuracy verify	no	no	no

2.3 Conclusion of Part I

The published literature on these cohorts reports discrimination that, where it exceeds the dataset authors’ baselines, coincides with day-level or unstated splitting; it does not report calibration; it does not test transfer between cohorts; and it rarely releases code. Under PROBAST+AI these are high-risk judgements in the analysis domain (signalling questions on optimism, calibration and sample size), and under TRIPOD+AI they are missing items 10d, 12, 16, 19 and 21 in the published numbering. [Final proportions with Wilson 95% CIs, PROBAST+AI domain distribution and TRIPOD+AI adherence per item to be inserted from the extraction workbook after the formal search; the figures and tables regenerate from part1_studies.csv.] Part II measures what these practices cost on the same data.

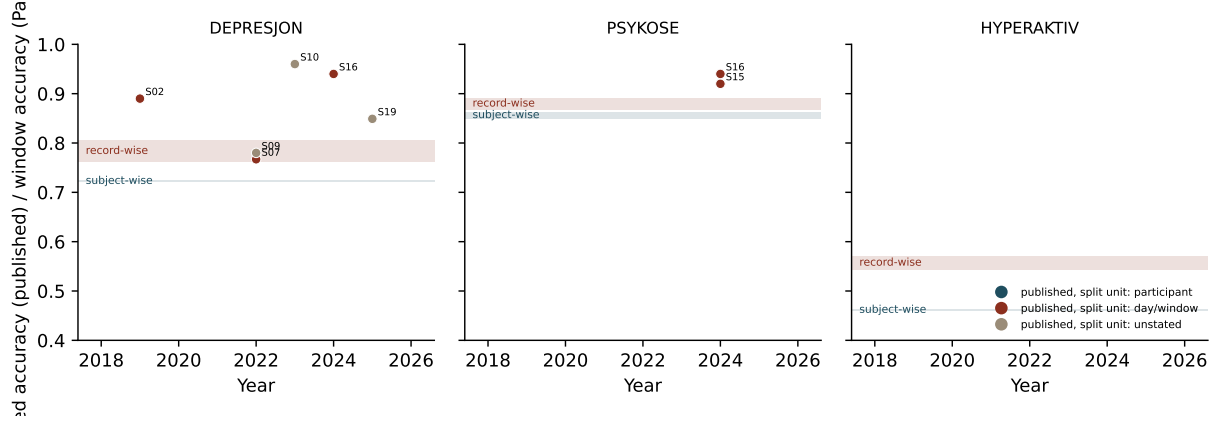


Figure 2: Published best accuracies (points, labelled by study ID in Table 1, coloured by the unit of splitting) against the window-level accuracy bands measured in Part II under subject-wise and record-wise cross-validation (shaded; range over logistic regression, random forest and XGBoost). Studies whose best metric is not accuracy, or whose value awaits full-text verification, are not plotted.

3 Part II: controlled re-evaluation

3.1 Data, participants and windows

Table 2: Participant characteristics per cohort. HYPERAKTIV controls are psychiatric patients without ADHD (clinical controls); DEPRESJON and PSYKOSE share the same 32 healthy controls. Sex, age band and recording days from the harmonised OBF-Psychiatric metadata; day-windows are the analysis units after the 80% validity threshold.

Cohort	Group	n	Female, n (%)	Age (modal band)	Recording days, median (range)	Valid day-windows	Clinical scale, median (IQR)
depresjon	cases (depression)	23	10 (43)	45-49	13 (5-18)	405	MADRS 24 (IQR 18-26)
depresjon	controls (healthy_control)	32	20 (62)	50-54	13 (8-20)	739	–
psykose	cases (schizophrenia)	22	3 (14)	35-39	13 (12-14)	390	BPRS 51 (IQR 44-58)
psykose	controls (healthy_control)	32	20 (62)	50-54	13 (8-20)	739	–
hyperaktiv	cases (adhd)	45	21 (47)	30-39	7 (2-10)	343	ASRS 49 (IQR 40-55)
hyperaktiv	controls (clinical_control)	40	20 (50)	17-29	7 (6-10)	329	ASRS 34 (IQR 25-43)
obf_psychiatric	cases (adhd, clinical, depression, schizophrenia)	130	54 (42)	30-39	7 (2-18)	1467	–
obf_psychiatric	controls (control)	32	20 (62)	50-54	13 (8-20)	739	–

DEPRESJON: 23 patients with unipolar or bipolar depression (MADRS at start median 24) and 32 healthy controls; PSYKOSE: 22 inpatients with schizophrenia (BPRS median 51) and the same 32 controls; HYPERAKTIV: 45 adults with ADHD and 40 clinical controls (psychiatric outpatients without ADHD) with activity recordings, read from OBF-Psychiatric. Hashing every participant’s activity series confirmed that the 32 PSYKOSE control recordings are byte-identical to the DEPRESJON control recordings and that OBF-Psychiatric contains each of the 162 participants exactly once. The analysis unit was one calendar day with at least 80% valid minutes; partial first and last days were dropped. DEPRESJON yielded 1,034 valid days (median 15 per participant), PSYKOSE 1,021 (16) and HYPERAKTIV 502 (6); 1,818 unique days; no participant was excluded (Table 2). Labels are participant-level diagnostic groups; the data schema rejects a participant with two labels.

3.2 Predictors, models and estimation

Twenty-eight predictors per day (Supplementary Table S2): distributional statistics, day/night contrasts, non-parametric circadian measures on the hourly profile (M10, L5, relative amplitude, intradaily variability, M10-onset phase), autocorrelation, spectral summaries, rest/activity transitions and missingness. No demographic variable, so that E2 compares populations rather than covariate availability. Models: majority-class baseline, L_2 logistic regression ($C = 1$, class-balanced),

random forest (500 trees, minimum leaf 2, balanced subsampling), XGBoost (300 rounds, depth 3, learning rate 0.05, subsampling 0.8), all fixed a priori and not tuned so that E1 isolates the splitting design; a three-block 1D-CNN on the raw $\log(1+x)$ minute series is a supplementary arm. Imputation (training-fold median) and standardisation are fitted on training folds only inside an `imblearn` pipeline; no resampling in the main runs. Metrics at *window* level (the unit most papers report) and at *participant* level (mean predicted probability per participant): AUROC, accuracy, balanced accuracy, macro-F1, MCC, Brier score, calibration slope and intercept [12], expected calibration error (ECE, ten bins) and reliability curves. Point estimates are computed on the seed-averaged out-of-fold predictions of ten seeds; 95% intervals are bias-corrected and accelerated bootstrap intervals from 1,000 resamples of *participants* with a leave-one-participant-out jackknife; per-seed means and SDs are reported alongside. Participant-level events per predictor (EPV) are the minority-class participants divided by 28.

3.3 Experiments

E1 leaky splitting. Five-fold cross-validation under a subject-wise splitter (stratified group k -fold; no participant on both sides of a fold) and a record-wise splitter (stratified k -fold over days), identical data and seeds; the paired difference, bootstrapped over the same participants, is the inflation. A test in the codebase fails the build if any configuration other than E1/E5 requests the record-wise splitter, and the runner asserts fold disjointness at run time. **E2 external validation.** Model fitted on all windows of cohort A and applied, frozen, to cohort B; internal comparator is subject-wise CV on A. Shared controls were assigned at random (seed 0) to exactly one cohort, 16 each, and the runner refuses any pair with a participant on both sides. **DEPRESJON** $\rightarrow$ **HYPERAKTIV** is a deliberate label-shift control. **E3 calibration.** All E1 (subject-wise) and E2 models, plus logistic recalibration (slope, intercept) fitted on the training cohort’s out-of-fold predictions and applied to the external predictions. **E4 sex strata.** Subject-wise E1 predictions stratified by sex from the OBF metadata. **E5 OBF-Psychiatric.** Any psychiatric group versus healthy control, and the five diagnostic groups (macro one-vs-rest AUROC), under both splitters. **E6 ablation.** Under subject-wise CV: drop-one and only-one feature groups (distributional, day/night, circadian, temporal/spectral), SMOTE inside training folds, and leave-one-subject-out in place of five-fold; logistic regression and XGBoost, five seeds.

3.4 E1: the cost of leaky splitting

Table 3: E1: discrimination under subject-wise versus record-wise cross-validation. Cells give the estimate on seed-averaged out-of-fold predictions with subject-level BCa 95% bootstrap intervals; inflation is the paired record-wise minus subject-wise difference.

Cohort	Model	Window AUROC subject-wise	Window AUROC record-wise	Window inflation	Subject AUROC subject-wise	Subject AUROC record-wise	Subject inflation
depresjon	Majority class	0.465 [0.292, 0.653]	0.491 [0.459, 0.528]	0.026 [-0.147, 0.218]	0.423 [0.300, 0.551]	0.427 [0.295, 0.555]	0.001 [-0.191, 0.202]
depresjon	Logistic regression	0.779 [0.675, 0.861]	0.845 [0.781, 0.896]	0.065 [0.024, 0.122]	0.899 [0.822, 0.969]	0.965 [0.927, 0.990]	0.065 [0.011, 0.109]
depresjon	Random forest	0.770 [0.681, 0.848]	0.877 [0.823, 0.923]	0.107 [0.058, 0.168]	0.883 [0.812, 0.948]	0.977 [0.949, 1.000]	0.094 [0.041, 0.146]
depresjon	XGBoost	0.764 [0.675, 0.842]	0.878 [0.830, 0.918]	0.114 [0.060, 0.181]	0.874 [0.795, 0.947]	0.985 [0.964, 1.000]	0.111 [0.040, 0.175]
psykose	Majority class	0.437 [0.267, 0.632]	0.500 [0.470, 0.535]	0.062 [-0.113, 0.241]	0.419 [0.278, 0.557]	0.462 [0.331, 0.593]	0.043 [-0.148, 0.227]
psykose	Logistic regression	0.907 [0.820, 0.942]	0.927 [0.875, 0.955]	0.020 [0.006, 0.041]	0.974 [0.943, 1.000]	0.994 [0.983, 1.000]	0.020 [0.000, 0.043]
psykose	Random forest	0.893 [0.798, 0.934]	0.937 [0.890, 0.962]	0.044 [0.017, 0.080]	0.956 [0.913, 0.997]	0.994 [0.983, 1.000]	0.038 [0.000, 0.074]
psykose	XGBoost	0.903 [0.810, 0.941]	0.945 [0.907, 0.969]	0.042 [0.016, 0.078]	0.969 [0.935, 1.000]	0.997 [0.991, 1.000]	0.028 [0.000, 0.059]
hyperaktiv	Majority class	0.493 [0.355, 0.628]	0.483 [0.433, 0.533]	-0.010 [-0.170, 0.148]	0.497 [0.398, 0.597]	0.443 [0.349, 0.540]	-0.054 [-0.203, 0.106]
hyperaktiv	Logistic regression	0.449 [0.360, 0.535]	0.579 [0.486, 0.661]	0.130 [0.095, 0.167]	0.424 [0.328, 0.519]	0.601 [0.502, 0.695]	0.177 [0.131, 0.221]
hyperaktiv	Random forest	0.443 [0.361, 0.534]	0.612 [0.526, 0.692]	0.169 [0.135, 0.206]	0.424 [0.333, 0.518]	0.641 [0.550, 0.722]	0.217 [0.177, 0.257]
hyperaktiv	XGBoost	0.442 [0.360, 0.524]	0.593 [0.518, 0.664]	0.151 [0.116, 0.186]	0.438 [0.337, 0.530]	0.652 [0.555, 0.732]	0.214 [0.173, 0.261]

Record-wise splitting raised window-level AUROC for every informative model in every cohort (Table 3, Figure 3). On DEPRESJON, XGBoost rose from 0.764 (95% CI 0.675 to 0.842) to 0.878 (0.830 to 0.918), a paired inflation of 0.114 (0.060 to 0.181); at participant level the leaky estimate reached 0.985. On PSYKOSE, where honest discrimination is already high (0.893 to 0.907), inflation was 0.020 to 0.044. On HYPERAKTIV the honest estimate was at or below chance for all models (0.442 to 0.449; participant level 0.424 to 0.438), yet record-wise splitting produced 0.579 to 0.612 at window level and 0.601 to 0.652 at participant level, an inflation of 0.130 to 0.169 with intervals

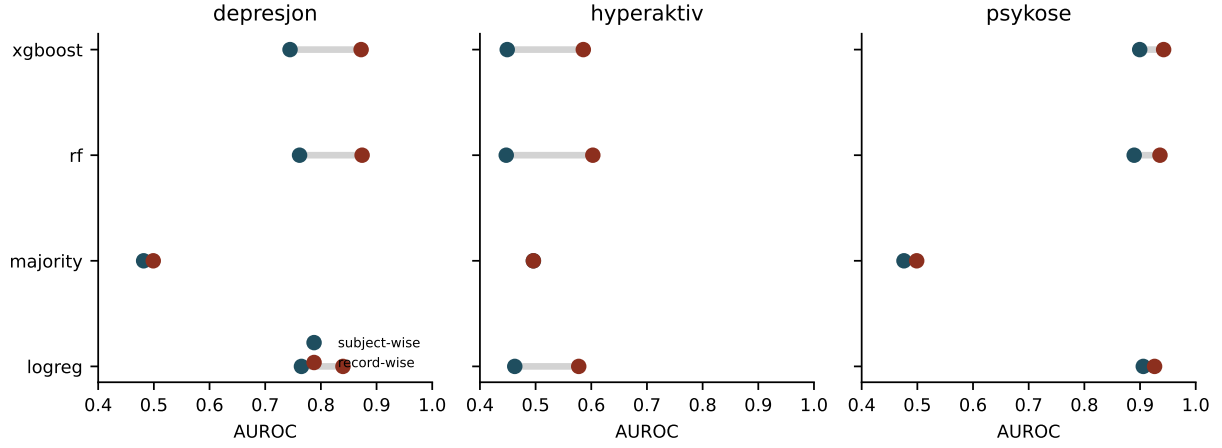


Figure 3: E1. Window-level AUROC of each model under subject-wise (dark) and record-wise (red) five-fold cross-validation; the grey bar is the leakage inflation.

177 excluding zero: leakage manufactured a signal. Logistic regression inflated least in every cohort
 178 (0.065, 0.020, 0.130) and the tree ensembles most. Accuracy and macro-F1 behaved identically
 179 (Table 4). The 1D-CNN did not exceed the feature-based models under honest splitting (0.696,
 180 0.849 and 0.459) and was unstable across seeds; its participant-level estimate rose from 0.738 to
 0.912 on DEPRESJON under record-wise splitting (Table 5), i.e. it memorised participants.

Table 4: E1: window-level discrimination and calibration for every model, cohort and splitting design (mean over seeds).

Cohort	Model	Split	Accuracy	Macro-F1	AUROC	Brier	Cal. slope	Cal. intercept	ECE
depresjon	Majority class	subject-wise	0.653	0.395	0.482	0.227	-4.01	-0.00	0.000
depresjon	Majority class	record-wise	0.653	0.395	0.499	0.227	0.29	0.00	0.000
depresjon	Logistic regression	subject-wise	0.722	0.706	0.765	0.194	0.52	-0.54	0.118
depresjon	Logistic regression	record-wise	0.761	0.750	0.840	0.163	0.90	-0.62	0.095
depresjon	Random forest	subject-wise	0.724	0.679	0.762	0.185	0.69	0.01	0.061
depresjon	Random forest	record-wise	0.806	0.782	0.874	0.136	1.23	-0.11	0.039
depresjon	XGBoost	subject-wise	0.722	0.676	0.745	0.202	0.43	0.29	0.123
depresjon	XGBoost	record-wise	0.802	0.776	0.872	0.137	0.82	0.07	0.039
psykose	Majority class	subject-wise	0.661	0.398	0.476	0.224	-3.95	0.00	0.000
psykose	Majority class	record-wise	0.661	0.398	0.499	0.224	0.31	0.00	0.000
psykose	Logistic regression	subject-wise	0.856	0.842	0.906	0.115	0.90	-0.60	0.086
psykose	Logistic regression	record-wise	0.868	0.855	0.926	0.103	1.07	-0.62	0.078
psykose	Random forest	subject-wise	0.850	0.829	0.889	0.118	0.95	-0.12	0.048
psykose	Random forest	record-wise	0.878	0.861	0.936	0.093	1.32	-0.17	0.048
psykose	XGBoost	subject-wise	0.862	0.842	0.899	0.108	0.71	0.15	0.046
psykose	XGBoost	record-wise	0.891	0.876	0.942	0.082	0.91	0.07	0.026
hyperaktiv	Majority class	subject-wise	0.504	0.335	0.496	0.250	0.00	0.00	0.000
hyperaktiv	Majority class	record-wise	0.504	0.335	0.496	0.250	0.00	0.00	0.000
hyperaktiv	Logistic regression	subject-wise	0.461	0.460	0.463	0.286	-0.21	-0.01	0.160
hyperaktiv	Logistic regression	record-wise	0.543	0.543	0.577	0.252	0.21	0.01	0.074
hyperaktiv	Random forest	subject-wise	0.463	0.462	0.447	0.286	-0.32	0.00	0.156
hyperaktiv	Random forest	record-wise	0.571	0.571	0.603	0.245	0.57	0.00	0.055
hyperaktiv	XGBoost	subject-wise	0.460	0.460	0.449	0.330	-0.15	0.00	0.248
hyperaktiv	XGBoost	record-wise	0.557	0.557	0.586	0.271	0.26	0.00	0.151

Table 5: E1: discrimination under subject-wise versus record-wise cross-validation. Cells give the estimate on seed-averaged out-of-fold predictions with subject-level BCa 95% bootstrap intervals; inflation is the paired record-wise minus subject-wise difference.

Cohort	Model	Window AUROC subject-wise	Window AUROC record-wise	Window inflation	Subject AUROC subject-wise	Subject AUROC record-wise	Subject inflation
depresjon	1D-CNN	0.696 [0.576, 0.795]	0.729 [0.671, 0.784]	0.033 [-0.063, 0.122]	0.738 [0.634, 0.844]	0.912 [0.837, 0.959]	0.174 [0.067, 0.270]
psykose	1D-CNN	0.849 [0.744, 0.910]	0.892 [0.836, 0.935]	0.043 [-0.000, 0.095]	0.922 [0.852, 0.976]	0.986 [0.963, 1.000]	0.064 [0.016, 0.111]
hyperaktiv	1D-CNN	0.459 [0.356, 0.588]	0.513 [0.447, 0.578]	0.054 [-0.079, 0.189]	0.455 [0.371, 0.563]	0.529 [0.432, 0.622]	0.074 [-0.060, 0.208]

Table 6: E2: internal (subject-wise CV on the training cohort) versus external (unchanged model applied to the second cohort) performance. Window-level estimates, mean over seeds, subject-level BCa 95% intervals. EPV = minority-class subjects per candidate predictor in the training cohort.

Train → Test	Model	EPV	Internal AUROC	External AUROC	Δ	Int. slope	Ext. slope	Ext. intercept	Ext. Brier
depresjon → psykose	Majority class	0.6	0.465 [0.224, 0.662]	0.500 [0.500, 0.500]	+0.035	-4.16	0.00	-0.02	0.250
depresjon → psykose	Logistic regression	0.6	0.779 [0.701, 0.871]	0.779 [0.676, 0.847]	-0.000	0.66	0.50	0.06	0.193
depresjon → psykose	Random forest	0.6	0.770 [0.607, 0.837]	0.776 [0.688, 0.846]	+0.006	0.58	0.80	-0.02	0.193
depresjon → psykose	XGBoost	0.6	0.764 [0.608, 0.825]	0.769 [0.678, 0.835]	+0.005	0.34	0.55	-0.01	0.204
psykose → depresjon	Majority class	0.6	0.390 [0.206, 0.601]	0.500 [0.500, 0.500]	+0.110	-4.07	0.00	0.06	0.250
psykose → depresjon	Logistic regression	0.6	0.885 [0.790, 0.938]	0.689 [0.564, 0.793]	-0.196	0.72	0.27	1.20	0.262
psykose → depresjon	Random forest	0.6	0.854 [0.756, 0.926]	0.728 [0.620, 0.818]	-0.126	0.76	0.63	0.74	0.225
psykose → depresjon	XGBoost	0.6	0.877 [0.793, 0.936]	0.671 [0.569, 0.770]	-0.206	0.55	0.30	1.44	0.285
depresjon → hyperaktiv	Majority class	0.8	0.465 [0.292, 0.653]	0.500 [0.500, 0.500]	+0.035	-4.01	-0.01	0.65	0.275
depresjon → hyperaktiv	Logistic regression	0.8	0.779 [0.675, 0.861]	0.442 [0.348, 0.532]	-0.338	0.52	-0.16	-0.42	0.350
depresjon → hyperaktiv	Random forest	0.8	0.770 [0.681, 0.848]	0.461 [0.360, 0.557]	-0.310	0.69	-0.15	0.33	0.317
depresjon → hyperaktiv	XGBoost	0.8	0.764 [0.675, 0.842]	0.459 [0.358, 0.550]	-0.305	0.43	-0.09	0.48	0.364

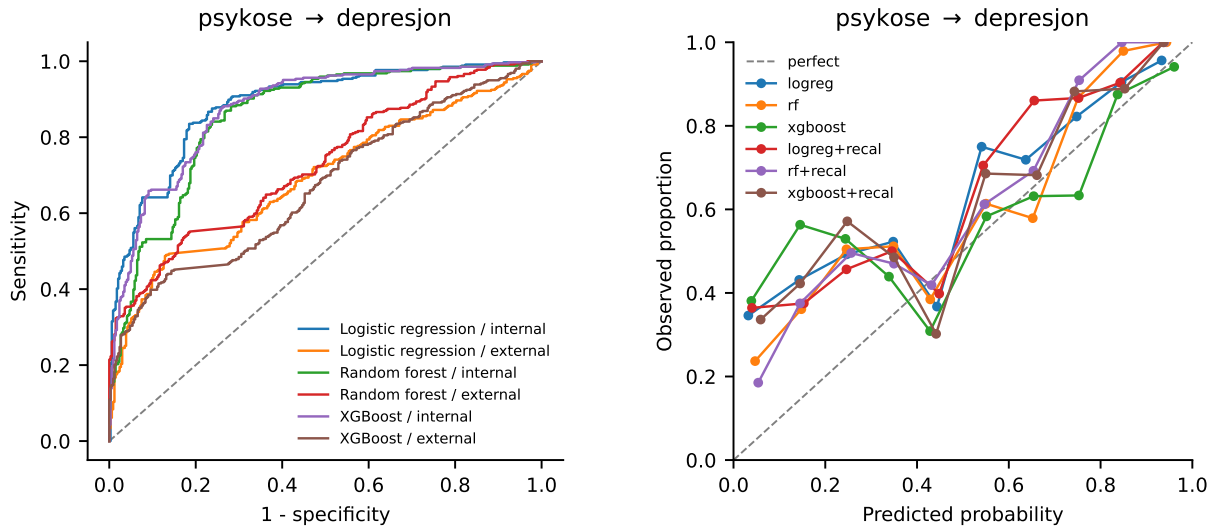


Figure 4: E2/E3, PSYKOSE → DEPRESJON. Left: ROC curves of internal (subject-wise CV on PSYKOSE) and external (frozen model on DEPRESJON) predictions. Right: reliability curves of the external predictions before and after logistic recalibration fitted on the training cohort.

Transfer was asymmetric (Table 6). DEPRESJON-trained models separated schizophrenia from the held-out controls as well as they had separated depression internally (external 0.769 to 0.779 versus internal 0.764 to 0.779), because schizophrenia produces the larger activity deficit. PSYKOSE-trained models lost 0.13 to 0.21 AUROC on DEPRESJON (logistic regression 0.885 → 0.689; XGBoost 0.877 → 0.671) and their probabilities were both overconfident and shifted: slopes 0.27 to 0.63, intercepts 0.74 to 1.44, Brier 0.22 to 0.29 against 0.25 for a constant prediction (Figure 4). The label-shift pair DEPRESJON → HYPERAKTIV fell to 0.442 to 0.460 with negative slopes. Participant-level EPV was 0.6 (DEPRESJON, PSYKOSE after the control split) and 0.8 (HYPERAKTIV).

3.6 E3: the calibration blind spot

Under honest splitting the informative models were already miscalibrated (Table 7, Figure 5): slopes 0.53 (XGBoost) to 0.77 (random forest) on DEPRESJON, 0.95 (logistic regression) on PSYKOSE, and negative slopes with ECE 0.14 to 0.22 on HYPERAKTIV, where the models had learned nothing yet produced confident probabilities. Record-wise models were markedly *better* calibrated in-sample, which is a symptom rather than a virtue. Recalibrating external predictions

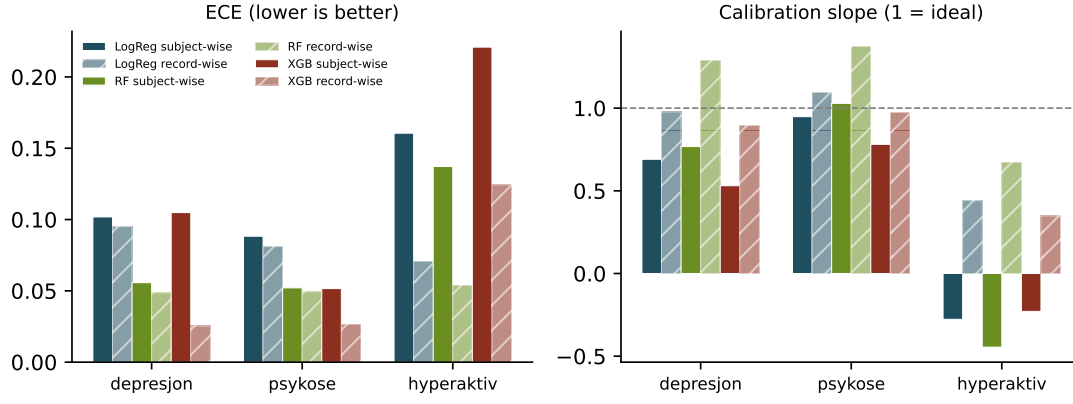


Figure 5: E3. Expected calibration error (left) and calibration slope (right) of every model under subject-wise (solid) and record-wise (hatched) cross-validation. Record-wise models appear better calibrated because they score participants they have seen.

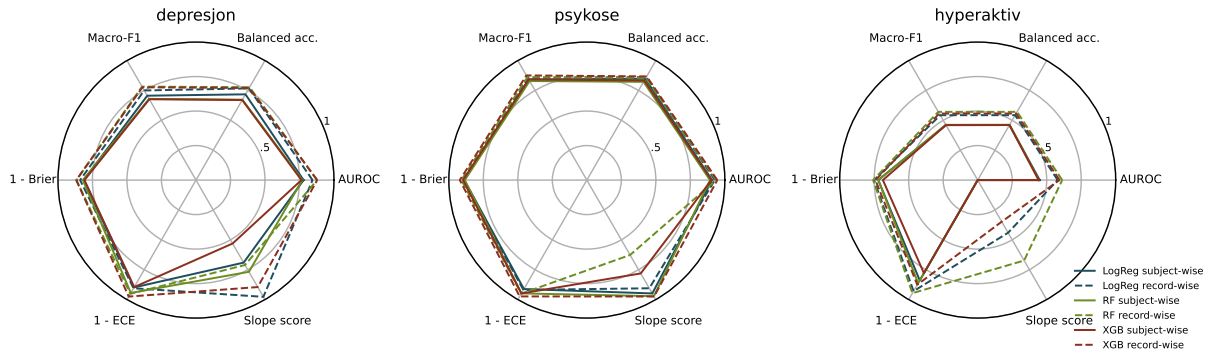


Figure 6: Multi-metric profile of the models per cohort (AUROC, balanced accuracy, macro-F1, 1–Brier, 1–ECE and a slope score $1 - |1 - \text{slope}|$) under subject-wise (solid) and record-wise (dashed) cross-validation. Leakage expands every axis at once; on HYPERAKTIV the honest profile collapses to chance while the leaky profile does not.

with a slope and intercept fitted on the training cohort repaired the easy direction (DEPRESJON → PSYKOSE: slopes 1.05, ECE 0.04 to 0.05 for the ensembles) but not the hard one (PSYKOSE → DEPRESJON: slopes 0.35 to 0.77, ECE 0.13 to 0.20), because the miscalibration there is a property of the population shift, not of the model.

Table 7: E3: calibration alongside discrimination for every model in E1 (subject-wise arm) and E2, plus logistic recalibration of external predictions.

Exp.	Run	AUROC	Brier	Cal. slope	Cal. intercept	ECE
E1	depresjon.logreg.subject_wise	0.779	0.187	0.69	-0.51	0.102
E1	depresjon.rf.subject_wise	0.770	0.181	0.77	0.01	0.056
E1	depresjon.xgboost.subject_wise	0.764	0.192	0.53	0.27	0.105
E1	hyperaktiv.logreg.subject_wise	0.449	0.279	-0.28	-0.01	0.161
E1	hyperaktiv.rf.subject_wise	0.443	0.281	-0.45	0.00	0.137
E1	hyperaktiv.xgboost.subject_wise	0.442	0.313	-0.23	0.00	0.221
E1	psykose.logreg.subject_wise	0.907	0.112	0.95	-0.59	0.088
E1	psykose.rf.subject_wise	0.893	0.115	1.03	-0.12	0.052
E1	psykose.xgboost.subject_wise	0.903	0.103	0.78	0.14	0.052
E2	depresjon-to-hyperaktiv.logreg.external	0.442	0.350	-0.16	-0.42	0.291
E2	depresjon-to-hyperaktiv.rf.external	0.461	0.317	-0.15	0.33	0.214
E2	depresjon-to-hyperaktiv.xgboost.external	0.459	0.363	-0.09	0.47	0.292
E2	depresjon-to-psykose.logreg.external	0.779	0.193	0.50	0.06	0.079
E2	depresjon-to-psykose.rf.external	0.776	0.193	0.81	-0.02	0.044
E2	depresjon-to-psykose.xgboost.external	0.769	0.202	0.56	-0.01	0.085
E2	depresjon.logreg.internal	0.779	0.187	0.69	-0.51	0.102
E2	depresjon.rf.internal	0.770	0.181	0.77	0.01	0.056
E2	depresjon.xgboost.internal	0.764	0.192	0.53	0.27	0.105
E2	psykose-to-depresjon.logreg.external	0.689	0.262	0.27	1.20	0.202
E2	psykose-to-depresjon.rf.external	0.728	0.225	0.63	0.74	0.149
E2	psykose-to-depresjon.xgboost.external	0.671	0.284	0.31	1.43	0.229
E2	psykose.logreg.internal	0.885	0.137	0.77	-0.10	0.061
E2	psykose.rf.internal	0.854	0.151	0.82	-0.08	0.055
E2	psykose.xgboost.internal	0.877	0.148	0.62	-0.16	0.083
E2	depresjon-to-hyperaktiv.logreg.external+recal	0.442	0.311	-0.23	0.22	0.223
E2	depresjon-to-hyperaktiv.rf.external+recal	0.461	0.301	-0.20	0.37	0.176
E2	depresjon-to-hyperaktiv.xgboost.external+recal	0.459	0.311	-0.17	0.36	0.194
E2	depresjon-to-psykose.logreg.external+recal	0.779	0.211	0.72	0.56	0.139
E2	depresjon-to-psykose.rf.external+recal	0.776	0.193	1.05	0.11	0.041
E2	depresjon-to-psykose.xgboost.external+recal	0.769	0.196	1.05	0.11	0.052
E2	psykose-to-depresjon.logreg.external+recal	0.689	0.254	0.35	0.99	0.185
E2	psykose-to-depresjon.rf.external+recal	0.728	0.221	0.77	0.67	0.129
E2	psykose-to-depresjon.xgboost.external+recal	0.671	0.258	0.49	0.93	0.200

3.7 E4 and E5: sex strata; transdiagnostic and five-class tasks

Discrimination did not differ materially by sex on DEPRESJON (0.76 to 0.78 in both strata) or HYPERAKTIV (chance in both); the PSYKOSE female stratum contains three cases and its apparently higher AUROC rests on them (Table 8). On OBF-Psychiatric, any psychiatric group versus healthy control reached 0.776 to 0.795 subject-wise and 0.824 to 0.849 record-wise (participant level 0.856 to 0.867 versus 0.899 to 0.943); the five-class task gave macro-AUROC 0.695 to 0.713 (accuracy 0.43 to 0.45 against a 0.28 majority rate) subject-wise and 0.770 to 0.811 (0.48 to 0.56) record-wise (Table 9), inflation again largest for the ensembles.

3.8 E6: ablation

[E6 paragraph is filled automatically from results/real/E6/e6_ablation.csv when the run completes; see Table ??.]

Table 8: E4: sex-stratified performance of the subject-wise E1 models (window level, subject-level bootstrap intervals). Subgroups with fewer than six participants or a single class are omitted.

Cohort	Model	Sex	Participants (cases)	AUROC [95\% CI]	Brier	Cal. slope	ECE
depresjon	Logistic regression	female	30 (10)	0.777 [0.649, 0.872]	0.182	0.75	0.120
depresjon	Logistic regression	male	25 (13)	0.776 [0.562, 0.882]	0.192	0.62	0.091
depresjon	Random forest	female	30 (10)	0.763 [0.623, 0.870]	0.171	0.75	0.071
depresjon	Random forest	male	25 (13)	0.777 [0.626, 0.870]	0.193	0.80	0.070
depresjon	XGBoost	female	30 (10)	0.755 [0.620, 0.859]	0.183	0.51	0.109
depresjon	XGBoost	male	25 (13)	0.772 [0.619, 0.868]	0.203	0.55	0.120
hyperaktiv	Logistic regression	female	41 (21)	0.479 [0.370, 0.602]	0.275	-0.10	0.152
hyperaktiv	Logistic regression	male	44 (24)	0.424 [0.321, 0.543]	0.283	-0.49	0.168
hyperaktiv	Random forest	female	41 (21)	0.491 [0.371, 0.611]	0.269	-0.10	0.111
hyperaktiv	Random forest	male	44 (24)	0.398 [0.278, 0.516]	0.291	-0.80	0.160
hyperaktiv	XGBoost	female	41 (21)	0.498 [0.385, 0.605]	0.293	-0.02	0.211
hyperaktiv	XGBoost	male	44 (24)	0.388 [0.278, 0.505]	0.332	-0.44	0.230
psykose	Logistic regression	female	23 (3)	0.971 [0.915, 0.998]	0.096	1.58	0.177
psykose	Logistic regression	male	31 (19)	0.890 [0.771, 0.933]	0.126	0.83	0.081
psykose	Random forest	female	23 (3)	0.966 [0.891, 0.995]	0.075	1.66	0.147
psykose	Random forest	male	31 (19)	0.874 [0.765, 0.930]	0.149	0.90	0.123
psykose	XGBoost	female	23 (3)	0.969 [0.904, 0.997]	0.059	1.16	0.100
psykose	XGBoost	male	31 (19)	0.884 [0.774, 0.933]	0.140	0.71	0.121

Table 9: E5: OBF-Psychiatric (162 participants, single control group). Transdiagnostic binary arm (any psychiatric group vs healthy control; window-level AUROC with subject-level BCa intervals) and five-class arm (macro one-vs-rest AUROC), each under subject-wise and record-wise CV.

Arm	Model	Split	AUROC / macro-AUROC	Accuracy	Macro-F1	Cal. slope
Any psychiatric vs control	Majority class	subject-wise	0.490 [0.358, 0.614]	0.641	0.391	0.25
Any psychiatric vs control	Majority class	record-wise	0.500 [0.470, 0.529]	0.641	0.391	0.25
Any psychiatric vs control	Logistic regression	subject-wise	0.795 [0.745, 0.848]	0.741	0.725	0.78
Any psychiatric vs control	Logistic regression	record-wise	0.824 [0.777, 0.867]	0.756	0.742	0.93
Any psychiatric vs control	Random forest	subject-wise	0.781 [0.726, 0.837]	0.739	0.705	0.75
Any psychiatric vs control	Random forest	record-wise	0.849 [0.811, 0.887]	0.782	0.755	1.07
Any psychiatric vs control	XGBoost	subject-wise	0.776 [0.729, 0.834]	0.734	0.702	0.63
Any psychiatric vs control	XGBoost	record-wise	0.844 [0.810, 0.885]	0.781	0.755	0.90
Five diagnostic groups	Logistic regression	subject-wise	0.713 [0.677, 0.750]	0.431	0.390	–
Five diagnostic groups	Logistic regression	record-wise	0.770 [0.740, 0.799]	0.484	0.444	–
Five diagnostic groups	Random forest	subject-wise	0.695 [0.662, 0.735]	0.434	0.350	–
Five diagnostic groups	Random forest	record-wise	0.811 [0.786, 0.843]	0.558	0.482	–
Five diagnostic groups	XGBoost	subject-wise	0.702 [0.670, 0.743]	0.448	0.355	–
Five diagnostic groups	XGBoost	record-wise	0.808 [0.785, 0.842]	0.558	0.467	–

3.9 Published claims against the measured bands

Table 10: Published best accuracies on each cohort (Part I seed list; Table 1) against the accuracy and AUROC bands measured in Part II under subject-wise and record-wise five-fold cross-validation (window level unless stated; seed-averaged estimates, range over logistic regression, random forest and XGBoost).

Cohort	Published best accuracies (ID value)	Part I accuracy, subject-wise (range over models)	Part II accuracy, record-wise	Part II AUROC, subject-wise	Part II AUROC, record-wise	Part II subject-level AUROC, subject-wise
DEPRESJON	S02 0.89, S07 0.77, S09 0.78, S10 0.96, S16 0.94, S19 0.85	0.72–0.72	0.77–0.81	0.76–0.78	0.84–0.88	0.87–0.90
PSYKOSE	S15 0.92, S16 0.94	0.86–0.87	0.87–0.89	0.89–0.91	0.93–0.95	0.96–0.97
HYPERAKTIV	–	0.44–0.46	0.53–0.57	0.44–0.45	0.58–0.61	0.42–0.44

Table 10 places the Part I claims beside the Part II bands. Every published accuracy above the subject-wise band on DEPRESJON (0.89, 0.96, 0.85) lies inside or above the record-wise band, and every such study used or did not state a day-level unit; the two PSYKOSE claims of 0.92 and 0.94 sit at the top of the record-wise band. The honest participant-level AUROC of 0.87 to 0.90 for depression agrees with the dataset authors’ leave-one-patient-out baseline and is what a new participant should expect; the leaky 0.97 to 0.99 is what the literature reports.

4 Discussion

4.1 Principal findings

Each practice the appraisal flags changed conclusions on the data, always towards optimism. Record-wise splitting added 0.02 to 0.17 AUROC and turned a null result into an apparently useful ADHD classifier. Skipping external validation hid a loss of up to 0.21 AUROC and a collapse of calibration in one transfer direction while the other direction happened to hold, which is why transfer must be tested rather than assumed. Calibration, unreported in the entire literature on these cohorts, was poor under honest validation and was not repairable from the training data alone. The honest results were robust to feature subsets, resampling inside folds and the cross-validation scheme (E6), so the gap between published and honest numbers is not a matter of feature engineering.

4.2 Why the combination matters

An appraisal alone counts practices; a re-analysis alone is one more model on familiar data. Together, on the same cohorts, they show that the practices the appraisal flags are the practices that produce the reported numbers. This is the strongest form of evidence a methodological critique can offer: not that a practice *could* inflate results, but that it *did*, by how much, and for which claims.

4.3 Limitations

One device family and one country; small participant numbers with wide participant-level intervals; diagnostic-group labels rather than symptom trajectories; one window length; models fixed rather than tuned (tuning inside subject-wise folds would raise honest performance somewhat without changing the paired inflation, which is the estimand); no demographic predictors; in E2 a control group split rather than an independent one, because no independent healthy-control actigraphy with the same device is public; the PSYKOSE sex analysis is underpowered; and Part I’s seed extraction relies partly on abstracts until the formal search and dual full-text extraction are completed [to be replaced by final counts].

4.4 Recommendations and conclusion

For any repeated-measures mental-health dataset: state the unit of splitting; report participant-level as well as window-level sample sizes and metrics; report calibration slope and intercept with a reliability curve; check for shared participants across combined datasets; release code with run manifests. Journals could require each in one sentence. This repository provides a reference implementation and honest baselines for the three cohorts; claims that exceed them should be accompanied by the unit of splitting that produced them.

Declarations

Funding: [none / grant]. Conflicts of interest: none declared. Ethics: secondary analysis of public, de-identified data; [institutional determination]. Registration: [PROSPERO/OSF]. Data: available from the dataset authors [1, 2, 3, 4]; SHA-256 checksums in the repository. Code: <https://github.com/rogerpanel/cv/tree/main/mlmh-appraisal> ([release tag; Zenodo DOI]). CRediT: A.A.A. conceptualisation, methodology, investigation, writing; R.N.A. software, validation, data curation, visualisation, writing.

Supplementary material

S1 TRIPOD+AI checklist; S2 feature definitions; S3 search log; S4 seed study sources; S5 per-window predictions and run manifests.

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